

One of the biggest things that the industry is focusing at is the next-generation process node. The EDA industry is working on developing tools for 22nm and 15nm processes. We are also partnering with companies to work on every aspect from manufacture to design.

Another challenge for chip designers is to ensure the maximum yield of 80-85 per cent. EDA companies are working towards yield-driven designs tools.

Power is the most important consideration in any design today. Designers want to go high in processing power but with a very small real estate that consumes little power. How to bring the power of desktop in your hand (smart-phone), how to create innovative power modules and finally how to design a green sub-system?

Embedded software is another area of focus. Development of tools for Android platform, cross-platform development and embedded Linux is gaining interest.

On the PCB side, companies are trying to provide comprehensive solutions for thermal management and place-and-route all in a single package.

Work continues on the emulation side. Developing more intelligent tools is what the industry aims at.

What are your views on multicore processors for embedded design?

Anything that happens in processors and graphics for high-end applications soon reflects in smaller embedded applications too. Multicore is already a reality. We have already seen people working on multicore processors. In the embedded market too, these are picking up. It's not only the chip side of it but also the software which will support the embedded side of it. The good news is these software are already available. So in the next 18-24 months, chips with multicore will find applications in networking and handsets.

There is a lot of talk about 3D chip designing. What's your take?

3D has just started. So it's too early to comment. But we hope that in the next three-four years, this will become another area which will provide designers the choice of packing more and more functionality in a given chip.

For manufacturing, we already have 3D IC checking, optical and process correction, and phase-shift mask on the Calibre platform. Apart from that, we are also working on tools to support 3D IC design.

You mentioned Open Source platforms like Android and embedded Linux. How do you see the acceptance of Open Source tools for design and development?

For anything to have wide adoption, Open standards are must. That makes it easier for the user to accept it psychologically. For something which is based on closed standard today, the adoption may be restricted. Open standard al-

ways help broaden the customer base.

What Mentor Graphics provides is a unique base or its own unique algorithm in Open standard. There is an Open standards body which designs the standards, but at the end, it boils down to individual implementation of these standards. This is where Mentor Graphics provides value.

If you look at the Open standard source code, it comes with a note saying that it is not production-proven. You have to do your own implementation to make it production-proven. That is the reason why Mentor has chosen to partner with customers or design houses. We help them understand how to take an Open Source platform and create an application to make it easier to productise it so that they can focus on their actual development and leave RTOS or software work to someone who knows it the best.

Is adoption of Open Source and Open standard seen as a threat in the proprietary world?

This is where we have to avoid the confusion. We are still using Open standards. The proprietary part comes in our software, our intelligence. Designers want it that way. Else, how will they distinguish? Designers want robustness, support and intelligence, so it's not either-or.

We are part of the ecosystem. Something which is proprietary is like Apple operating system. But in the case of RedHat Linux, it is still using Open Source. Similarly, Mentor is providing Linux and Android tools. Support and infrastructure have a very symbiotic relationship in the entire process.

How do you find the pricing of EDA tools? People are ready to pay for hardware tools but investing money on software is still not an accepted concept, which leads to piracy.

We are working with our clients on pricing. We are working with them on models like start-up model where we help small companies incubate. This is an innovative model to take them off the ground.

Piracy is a very sad thing. There is not much we can do about it. Piracy is a bit more common in Asian countries than in the US and other countries. We try to work with those companies, giving them innovative business ideas. Piracy is a problem that the entire industry faces and therefore we have also formed an anti-piracy group to address individual cases.

What kind of support do you provide to universities?

Universities are a major part of the ecosystem. Recently, we tied up with JNU and donated tools for their VLSI labs. We are closely working with Trident to provide universities with tools and support they need, as it's not only about tools but how to use them and how to use them effectively.

We are working with many more technical institutes to develop designers' community.